



NAME	ROLE	EMAIL
Bagnoli Antonio	MA	antonio1.bagnoli@mail.polimi.it
Bandello Federico	GNC	federico.bandello@mail.polimi.it
Bergamin Chiara	STR	chiara.bergamin@mail.polimi.it
Bernasconi Ludovico	TCS	ludovico.bernasconi@mail.polimi.it
Chico Alfonso Santiago	ADCS	santiago.chico@mail.polimi.it
Davì Marco	EPS & OBDH	marco.davi@mail.polimi.it
Emrem Mert	ROB & MECH	mert.emrem@mail.polimi.it
Pozzoni Anna Franca	COST, RISK & PROGR	annafranca.pozzoni@mail.polimi.it
Sagliani Lorenzo	TMTC	lorenzo1.sagliani@mail.polimi.it
Troggio Daniele	PROP	daniele.troggio@mail.polimi.it

ACADEMIC REFERENCE

COURSE:	Applied Space Mission Analysis and Design
A.Y.:	2025/2026
PROFESSOR:	Prof. Michèle R. Lavagna
STUDY PHASE:	Phase A study

1 MISSION OVERVIEW

1.1 Introduction

As humanity pivots from lunar exploration towards a sustained **Martian presence**, the orbital environment of the Red Planet is undergoing a radical transformation. The deployment of **PNT and relay constellations** marks the transition to a permanent infrastructure that cannot rely on Earth-based replacement cycles. **PHAROS-M** (Planetary Hub for Assistance, Refueling & Orbital Servicing around Mars) is a strategic response to this challenge, redefining Martian operations through **In-Situ Resource Utilization (ISRU)** and orbital sustainability. By providing autonomous **refueling, relocation, and repair** services, this mission ensures that the critical assets of today become the enduring foundations of a multi-planetary future. PHAROS-M aims to support the needs of future missions as an essential part of future Mars infrastructure.

1.2 High Level Objectives

The primary goal of **PHAROS-M** is to deploy a dedicated space segment capable of performing On-Orbit Servicing (OOS) for multiple assets in the Martian environment. The mission is driven by three core objectives: **Operational Continuity**, i.e., providing refueling and servicing capabilities to **extend the operational lifetime**; **Mission Flexibility**, i.e., enhancing the versatility of the Martian orbital network by enabling assets to be repurposed or sustained beyond their original design life; and **Infrastructure Sustainability**, i.e., ensuring the **sustained functionality** of critical nodes around Mars, reducing the need for replacement launches and mitigating the risk of orbital obsolescence.

1.3 High Level Requirements

To guide the design and operational scope of the **PHAROS-M** mission, the following high-level objectives and system requirements have been defined:

- **Transportation:** The servicer shall be transferred to the Martian environment by a dedicated Martian carrier.
- **Communications:** The servicer should benefit from the Martian carrier's relay service, if required for mission operations.
- **Resource Utilization:** If needed, the servicer shall consider the resources available within the Martian environment (whether stored on the surface or in orbit).

- **Servicing Capabilities:** The mission design shall support relocation, refueling, and repairing operations.
- **Multi-Client Capacity:** The servicer shall support servicing of **at least three satellites** using a single on-board propellant load.
- **Operational Lifetime:** The servicer shall have a lifetime at least equal to the mean lifetime of science spacecraft currently operating at Mars.
- **Technology Sourcing:** Preference should be given to the utilization of European technologies throughout the system design.

2 DRIVERS AND CRITICALITIES

The mission concept is driven by the need to **interact with a heterogeneous client population**, encompassing varying **degrees of cooperation** and **servicing-readiness by design**. Beyond this, the servicing needs are inherently non-uniform: candidate clients exhibit different masses and inertia properties, remaining lifetimes, and propellant consumption rates, and may require distinct intervention types depending on their onboard architecture. In addition, clients can present different degrees of autonomous capability, ranging from fully cooperative spacecraft able to support rendezvous and attitude stabilization, to partially cooperative or degraded spacecraft with limited propulsion and attitude-control authority. In parallel, some **criticalities** drive the overall solution space, notably the targeted autonomy level for Mars operations, the selected communication strategy, strict mass limitations imposed by the carrier, and the requirement to preserve mission flexibility in terms of ΔV and reachable clients.

3 DESIGN PHILOSOPHY

The PHAROS-M architecture is driven by a service-oriented philosophy designed to maximize accessibility to the Martian customer base. A statistical analysis of the current and projected orbital population identifies a specific **high-density operational cluster** within the **high-inclination regime**, where the majority of scientific and telecommunication assets are concentrated.

Consequently, the platform is strategically sized to operate across this wide orbital envelope without requiring any design modification.

This versatility minimizes the propellant (ΔV) required for phasing and rendezvous maneuvers while maximizing the frequency of servicing windows, effectively positioning PHAROS-M as the central node of the developing Martian orbital infrastructure.

4 ENVIRONMENT

Regarding the Martian environment, it is necessary to consider that the large Earth-Mars distance entails significant communication latencies (from 3 to 22 minutes) and logistical constraints, driving the need for high system autonomy. Furthermore, the solar fluxes are strictly dependent on the orbital position due to the Martian eccentricity. Consequently, the external loads vary significantly between 590 W/m^2 (Aphelion) and 717 W/m^2 (Perihelion).

Regarding radiation, Mars lacks a global magnetic field, so no trapped particles and no magnetic shielding are present. Solar particles and Cosmic Galactic rays are the main source of radiation. Using the SPENVIS software, a minimum **total ionizing dose (TID)** hardness requirement for all the avionics is set: $\geq 30 \text{ krad(Si) @30 mm Al}$. Moreover, the mission is potentially subject to high Single-Event Upset rates that needs to be mitigated by implementing appropriate redundancy, shielding and error-correction techniques.

5 ARCHITECTURE

Based on these drivers and criticalities, a structured trade-off based on the **Analytic Hierarchy Process (AHP)** led to a baseline centred on a **single-vehicle servicing architecture** enabling repeated operations within a **single orbital cluster**. The selected concept relies on **relay-based communications** to support autonomous operations around Mars and integrates a **robotic arm** and a **standardised refueling interface**, ensuring compatibility with diverse clients while keeping operational complexity, risk and resource margins within acceptable bounds.

5.1 Servicing Tasks and Operations

The servicing tasks of the missions are comprised of refueling, relocation, and repair. The repairing tasks identified as feasible within the scope of the mission are: **replacement** of modular critical units (such as valves, on-board computers, sensors and batteries) and **Orbital Replacement Units (ORUS)**, **maintenance** or replacement of solar arrays and thermal blankets,

assisted deployment of antennas and solar arrays.

All servicing tasks are to be carried out **fully autonomously**, which is motivated by the communication delay and the environmental variability.

5.2 Space Segment Overview

As previously mentioned, **PHAROS-M** will reach Mars thanks to the **Martian carrier**. Therefore, the satellite's stowed configuration must respect the geometrical constraints imposed by the carrier: 3.15 m maximum lateral dimension, 4.48 m maximum diagonal dimension, and 3.24 m maximum height. As can be seen from the stowed and deployed configurations presented in Figure 1 and Figure 2, the stowed configuration is compliant with the requirements.

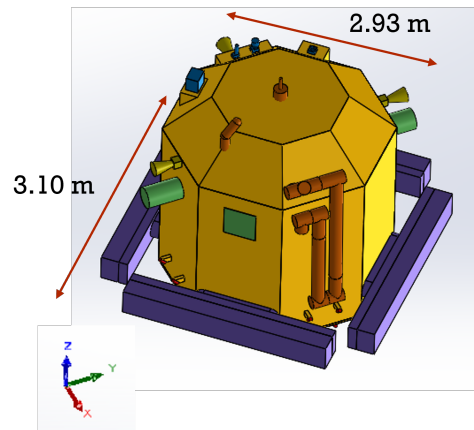


Figure 1: **PHAROS-M** folded configuration

To have a reference on the deployment of the **ROSA** panels and of the **Robotic Arm**, the fully deployed configuration is shown, where the maximum diagonal dimension of 13.62 m is obtained from the farthest corner of one array from the bus, to the one of the opposite panel.

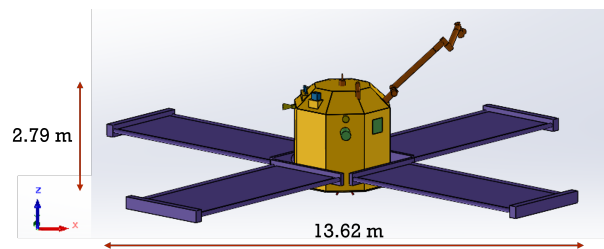


Figure 2: **PHAROS-M** deployed configuration

In Table 1, the main dimensions and mass properties of **PHAROS-M** are presented:

Property		Value
Bus	Dimensions	2.93 × 2.79 m ($W \times H$)
Solar Array (ROSA)	Size ($W \times L$)	2.30 × 5.46 m
Dry Mass		1279.43 kg
Propellant (Xe + Ar)	Mass	234.76 kg
Cables and Har- nesses		49.21 kg
Wet Mass (Launch)		1563.40 kg

Table 1: Physical and Mass Properties

The dimension of 2.93 m refers to the maximum lateral dimension of the octagonal bus; while 2.79 m is evaluated from the ring adapter to the top part of the refueling port. The **solar panels** dimensions refer to their global size, not only the power generation part.

6 CONCEPT OF OPERATIONS

The **PHAROS-M Concept of Operations (ConOps)** is defined as a mission **timeline** composed of an initial deployment and commissioning phase, followed by a sequence of repeated **servicing scenarios** around Mars. After release from the carrier and transfer to the operational region, PHAROS-M establishes a reference operating orbit and executes a recurring campaign for each target: **client selection and planning**, transfer to the client orbit, rendezvous, servicing (**refueling, relocation, repair**), and post-servicing separation and return to the reference orbit. This cycle is repeated for multiple clients over the mission lifetime, and the timeline closes with **end-of-life** passivation and disposal activities according to the selected end-state strategy.

7 SUBSYSTEMS OVERVIEW

7.1 Mission Analysis

The **PHAROS-M** MA subsystem establishes the orbital architecture and the trajectory design framework that enable repeated servicing operations around Mars. The mission concept is organized in a cluster-driven logic, where a representative **parking orbit** is defined as a recurrent operational hub and each servicing campaign is modeled as a sequence of **round-trip** legs between the parking orbit and the client orbit. Within this framework, MA consolidates a unified propulsion philosophy by separating **deter-**

ministic contributions, associated with planned transfers and mission phase changes such as injection and disposal, from **stochastic** allocations, introduced to preserve feasibility under operational dispersions and long-duration maintenance needs. This structure emphasizes that the overall propellant demand is primarily transfer-driven, while uncertainty and variability are absorbed through margins.

The reference operational scenario and the corresponding cluster are selected through a dedicated trade-off that remains **interchangeable** across all candidate clusters. A parking orbit is chosen to minimize the transfer cost to the cluster orbital range and to remain consistent with current and planned mission populations. A **parameter of merit** is then adopted to discriminate among clusters by combining the number of reachable clients with a **saturation ratio** that reflects the relative demand for servicing within the overall Mars orbiter population. This approach identifies the preferred design cluster while avoiding solutions dominated by orbit outliers, ensuring that the final baseline remains representative and extensible to alternative operational regions if mission priorities change.

Beyond architecture selection, MA characterizes the **operational envelope** by quantifying which final orbits are reachable from the parking orbit within the available low-thrust capability. The envelope is derived by mapping the transfer cost in the space of semi-major axis, eccentricity, and inclination, generating **reachable iso-surfaces** that link a per-leg cost to a guaranteed serviceable population. Under the current baseline allocation, the reachability assessment supports a **guaranteed capability to service at least five clients**, while higher serviceable counts correspond to progressively more restrictive reachable domains.

Trajectory generation and verification adopt a deterministic low-thrust baseline based on **Q-law**, used as a robust steering policy to drive the orbital elements toward the target set. Thrust execution is explicitly constrained by **power availability**: during eclipse periods, the electrical thruster is inhibited, and transfers are propagated through thrust arcs in sunlight only, ensuring consistency with the subsystem power balance. Operational timing is treated under a planning-oriented philosophy. Large plane-change corrections are avoided whenever possible by exploiting **conjunction windows** between orbital planes, whose evolution is predicted through **RAAN precession under the J_2 effect**. This implies that mission responsiveness is not the primary driver, while the transfer start time and duration are required to remain

compatible with the conjunction opportunities. A dedicated **phasing strategy** is employed to close the relative phase with the target and enable reliable rendezvous timing, and a perturbation-aware **station keeping** strategy with dead-band and hysteresis logic limits corrective actions to threshold exceedance, reducing duty cycle and preventing chattering.

7.2 Guidance, Navigation & Control

The **PHAROS-M** GNC subsystem is designed to support fully autonomous rendezvous and docking operations in the Martian environment, limiting ground intervention to critical authorization gates.

During nominal operations, phasing, and early rendezvous, absolute navigation relies on **ground-based orbit determination**. The on-board transponder facilitates two-way Range and Doppler tracking via the **MANTIS** relay orbiter. These measurements are downlinked to Earth for processing, and the updated absolute state vector is subsequently uploaded to PHAROS-M via MANTIS during the next visibility window. This loop sustains the approach until the **Narrow-FOV Camera** achieves valid target detection - approximately at 20 km, depending on the client's size and albedo - triggering the handover from absolute to relative navigation.

The relative state is estimated by an **Extended Kalman Filter (EKF)** that fuses multi-spectral data based on the operational range. In the Far-Range (from 20 km down to 1.5 km), the Narrow-FOV Camera provides angles-only bearing data. Upon entering the **Approach Sphere** ($R < 1.5$ km), the system transitions to Proximity Operations, engaging the full sensor suite - comprising a **Wide-FOV Camera**, **IR Camera**, and **3D LiDAR** - for **6-DOF pose estimation**. This suite is aided by **star trackers (ST)** and high-precision **Inertial Measurement Units (IMU)**, which provide essential attitude knowledge and state propagation between updates. This sensor fusion ensures robustness against difficult lighting conditions, such as eclipse or sun-blinding, and meets the strict docking requirement ($AK E \leq 0.01^\circ$).

The baseline guidance strategy adopts a **Radial-Radial (R-R) V-bar hopping** scheme. While this approach is less propellant-efficient than tangential methods, R-R impulses were selected to guarantee **passive safety** by maintaining bounded relative motion (with no semi-major axis change) in the event of missed burns. The approach profile is constrained by an 8° half-angle corridor to ensure continuous sensor visibility. **Safe Hold Points (SHPs)** are established at -1.5 km (First Hold) and -50 m (Last Hold),

where explicit **Ground-in-the-Loop** authorization is mandatory to proceed. Between these gates, intermediate hold points are placed using a **Non-Linear Programming (NLP)** optimization to maximize corridor clearance. Inside the **Keep-Out Zone** ($R < 50$ m), the GNC subsystem supports both inspection maneuvers and final approach profiles for docking ports located at various orientations (Zenith, Nadir, or Velocity). Once undocked, a specific Release maneuver transfers PHAROS-M to a **Safe Retreat Point** located at -550 m along the V-bar (500 m from the last hold).

Regarding control stability, sensitivity analyses using both **Linear Covariance (LinCov)** and **Monte Carlo (MC)** simulations - accounting for differential disturbances with the client - indicate that open-loop drift at hold points exceeds safety margins over long durations. Consequently, an **active relative station-keeping** strategy based on LQR logic is implemented, allocating approximately 2.8 m/s for a 5-client campaign. The total ΔV budget for Cluster 2 proximity operations is estimated at **9.5 m/s**. This budget is sized for Cluster 2 **proximity operations over 5 clients**, covering the full Rendezvous and Proximity Operations (RPOs) needs, including **Collision Avoidance Maneuver (CAM)**. This requirement corresponds to **15.10 kg** of Argon for the secondary propulsion system. The total operational timeline is capped at **19.5 hours** from the first hold to the safe retreat point; this duration accounts for 30-minute calibration and filter convergence stops at each hold but excludes the variable wait times for Ground-in-the-Loop authorization, which are dictated by **MANTIS** relay contact windows.

7.3 Attitude, Determination & Control Subsystem

The **PHAROS-M** ADCS provides full **3-axis attitude determination and control** across all mission phases, in both **standalone-servicer** and **docked-stack** configurations. The subsystem is driven by a mission-wide **pointing and knowledge budget** that captures the **absolute-attitude** requirements imposed by power generation, communications and thrust-vector alignment, and extends to the **relative-attitude** needs of RPOs, docking and berthing. A dedicated **safe Sun-pointing** capability enables autonomous stabilization after major anomalies, maintaining a thermally- and power-safe attitude until ground intervention.

This required pointing/knowledge envelope is closed through an **ECSS/ESA pointing-error budget** approach, combining statistically inde-

pendent Pointing Error Sources to obtain 1σ values and enabling **3σ compliance** verification. The resulting closure remains coherent across the full mode set, from absolute attitudes to the most demanding relative-attitude phases supporting RPO and servicing.

The ADCS **architecture** implements a **redundancy** philosophy aligned with mode criticality. Absolute attitude determination is driven by **2+1 star trackers** (cold redundancy) and **2 IMUs** (hot redundancy) to ensure continuous rate sensing and robust propagation between attitude updates. Safe and Sun-pointing modes are supported by a distributed Sun-sensor set (**8 FSS + 8 CSS**) providing functional redundancy through geometric coverage. Fine control and disturbance rejection rely on a **4+1 pyramidal reaction wheel** set (cold redundant wheel), while **16 resistojets** provide **momentum unloading** and additional torque authority for off-nominalities.

Detumbling is achieved through the control authority provided by the resistojets, enabling a **ω -vector rate damping** after carrier release and after capture of an uncontrolled client. A 100% margined propellant mass of **1.83 kg** is allocated for detumbling needs.

ADCS performance is verified against configuration-aware sizing envelopes that capture servicer propellant depletion and the variability introduced by client attachment. **Slew capability** is assessed for two representative maneuver families: (i) **rate-limited** 180° reorientations at the structural-/mechanism-driven cap ($0.5^\circ/\text{s}$ in standalone, $0.15^\circ/\text{s}$ in stack), and (ii) **time-limited** inspection maneuvers (360° within one orbital period) representative of synchronized RPO attitude maneuvers. In **standalone** configuration, required maneuvers are achievable with adequate torque and momentum margins when all four wheels are available; under **single-wheel failure**, inspection remains feasible within its time constraint while the maximum-rate 180° reorientation cannot always be maintained without saturation, but achievable rates remain above **50%** of the requirement. In **stack** configuration, slew performance is **highly client-dependent** and primarily up-bounded by the client-to-servicer **inertia ratio** (IR): $IR \leq 1.5$ about X/Y axes and $IR \leq 4$ about Z. Beyond these limits, commanded slew rates must be relaxed to preserve RW margins and prevent saturation.

Long-duration **attitude station keeping** is ensured by continuous disturbance rejection with reaction wheels. The **disturbance environment** is analysed across all mission phases and attitude profiles; for each mode, external disturbance torques are projected onto the body axes to quantify the resulting RW **momentum build-**

up, which drives wheel saturation and defines the required desaturation strategy. **Desaturation** demand concentrates in long-duration attitudes (notably Parking) and in client Relocation (where 3-axis control is strictly required due to stack instability), while proximity phases are achievable with continuous RW authority, with no desaturation required throughout a RPO leg. The consolidated desaturation propellant allocation (including a 100% margin) is **5.96 kg**, supporting an operational concept where momentum dumps are scheduled by phase and performed ahead of large slews when required.

7.4 Propulsion Subsystem

The primary propulsion system is designed to satisfy the most demanding mission ΔV requirements, selected to maximize the number of serviceable clients, ensuring nominally a **Total Impulse of 3.47 MNs**. The baseline solution relies on a **magnetically shielded 1.5 kW hall effect thruster**, qualified for over 20,000 hours of operation. Performance analysis confirms a specific impulse (I_{sp}) of **1725.7 s** and a thrust of **89.7 mN** using **xenon** as the baseline propellant for the nominal mission lifetime. To guarantee operational continuity, the system implements a **1+1 redundancy scheme** with a backup thruster unit held in **cold redundancy**. Capitalizing on the potential for In-Situ Resource Utilization (ISRU), the architecture adopts a sequential propellant strategy. Xenon remains the baseline for the nominal mission, utilized until depletion to maximize performance. Subsequently, the system allows for a transition to **argon** to support life-extension phases. Since standard Hall thrusters exhibit reduced efficiency with argon, the design proposes an **alternative configuration featuring an elongated discharge channel** to optimize ionization efficiency ($\eta \approx 40\%$), pending experimental verification before the Preliminary Design Review.

The secondary system handles ADCS assistance, station keeping, and high-precision control during proximity operations. The selected solution comprises **16 Resistojets**, delivering a combined **total Impulse of 30.47 kNs**. In nominal **hot mode** (95 W power consumption), these thrusters provide **250 mN** of thrust with an I_{sp} of **110 s** using **Argon**. To ensure immediate response capability, half of the assembly operates in **hot redundancy**. A key feature of this subsystem is its ability to operate in **cold gas mode**: this operational flexibility provides the ultra-fine impulse resolution necessary for safe and precise docking maneuvers.

The propellant storage architecture inte-

grates **four tanks of 60 L each**, arranged symmetrically to maintain a stable Center of Mass throughout the propellant depletion profile. Two tanks are dedicated to Xenon and two to Argon, with all vessels featuring a **strategic oversized volume design**. Regarding the Xenon tanks, this configuration accommodates the current baseline launch load of **205.85 kg**, but is specifically sized to **allow for a higher propellant load**, offering the flexibility to further extend the operational lifetime. Simultaneously, while the Argon tanks carry an initial mass of **28.91 kg**, their excess volume is dimensioned to **maximize storage capacity** for future in-orbit refueling operations, fully exploiting the ISRU strategy.

The servicing payload features a dedicated tank containing **55 kg of Hydrazine** for client refueling. The system employs a **helium blow-down architecture** designed with a final pressure constraint of **10 bar**, ensuring the capability to transfer propellant effectively even to partially filled client tanks.

System validation is performed using acausal **Simscape** models and a comprehensive **Monte Carlo analysis**. Results confirm that the architecture and thrust levels satisfy all mission constraints, demonstrating robustness against thermal uncertainties and power consumption variations.

7.5 Telemetry and Telecommand subsystem

PHAROS-M communicates with the Ground Segment at Earth by exploiting a relay link provided by **MANTIS**, the carrier mission. This choice allows to save mass and power and to reduce the complexity of the spacecraft configuration by avoiding the use of large antennas. The mission is designed to function by relying on **MAN-TIS** only, but also other future Martian relay providers are identified as a backup, such as **ESA's MARs Communication and Navigation Infrastructure** and **Blue Origin's Mars Telecommunications Orbiter**.

The on-board Telemetry and Telecommand subsystem operates in the **UHF** band, in the specific frequency range allocated by the SFCG Recommendation 22-1R4 (390-405 MHz and 435-450 MHz). Two **Electra UHF transceivers** (NASA JPL, TRL 9) are mounted on board, with one of them being in cold redundancy. BPSK modulation and Convolutional 1/2 encoding are implemented, as supported by the CCSDS UHF Proximity-1 protocol. Coherent two-way range and Doppler is supported for the tracking performed by **MANTIS**. Downlink is performed using a compact 8.7 dB gain **helix antenna**. An ad-

ditional identical antenna is mounted in cold redundancy. Uplink is received by two low-gain **patch antennas**, mounted on opposite sides of the spacecraft to maximize coverage. The RF chain architecture is designed such that every antenna can be used both for transmission and reception, in case it is needed in non-nominal scenarios. At the maximum distance from **MAN-TIS**, the system can achieve a maximum data rate of 0.46 Mbit/s in downlink and 8 kbit/s in uplink.

Because of the communications delay experienced at Earth-Mars distances, the mission shall operate with a high level of autonomy. The communication windows between **PHAROS-M** and Earth are exploited as check-points to assess spacecraft's nominality and to authorize the execution of the next autonomous/automatic set of operations. Moreover, high-resolution images of the client can be acquired and transmitted to Earth to support the preparation of Capture and Servicing operations from Ground.

7.6 On-Board Data Handling

The **On-Board Data Handling (OBDH)** subsystem is responsible for command execution, platform data management, payload data handling, on-board storage and processing. The mission presents **moderate storage needs**, as the dominant information flow to Ground consists of housekeeping and health telemetry, while payload data are concentrated in specific operations. In particular, **Approach & Capture** and selected servicing phases require the management of high-resolution client imagery and robotics/vision data to support ground-assisted planning and verification activities.

The OBDH architecture is conceived to support **high autonomy and robustness** in a long-duration Mars orbital mission. A **single-fault-tolerant baseline** is adopted through a **dual-string Command and Data Management Unit (CDMU)** with automatic or ground-commanded switchover governed by **FDIR**. A **two-layer communication strategy** is implemented to separate deterministic low-rate traffic from high-rate payload data handling: **CAN** is used for command and housekeeping exchanges with platform units, while **SpaceWire** provides a dedicated high-throughput path for payload/vision data and internal mass-memory interfaces. A dedicated **Robotics (& Vision) Computing Unit (RCU)** is included to enable local compute capability and to host safety-critical and update-sensitive logic, with degraded-mode fallback on the CDMU to preserve safe operations in case of anomalies. On the other hand, **ADCS Interface**

Unit (AIU) and **Remote Terminal Unit (RTU)** are in charge of the platform/housekeeping and attitude components management.

Computing throughput and memory requirements were derived using a **data-driven sizing workflow**. Processed data streams were converted into **32-bit word rates** and mapped into CPU workload through an instructions-per-word model, while buffering times and data handling logic determined the associated **RAM** needs; the **ROM** requirement follows from the software code footprint (platform services, FDIR, and mission applications). For payload units integrating on-board FPGA processing (e.g., smart cameras and navigation sensors), sizing is driven by the processed sensor output; for raw-output sensors, the integration of external FPGA pre-processing is assumed to **limit the effective data rate** entering the CPU, preventing unbounded growth of the processing load. A phase-dependent screening across mission modes then identifies the **worst-case operational envelope** for dimensioning, which is dominated by high-rate vision and relative-navigation activity during capture operations.

7.7 Electrical Power Subsystem

The **Electrical Power Subsystem (EPS)** of **PHAROS-M** is designed to ensure reliable power generation, storage, regulation, and distribution throughout all mission phases, with particular stress on transfer operations and long-duration Mars orbiting. Two main drivers shape the subsystem design: **stringent carrier mass and stowage constraints**, which impose a strong focus on high specific power and compact packaging, and **high peak power demand** in the most power-intensive modes, especially transfer phases dominated by electric propulsion. In addition, the mission lifetime in a Mars orbital environment implies frequent eclipse-driven charge/discharge cycling, which sets robustness and battery lifetime as key design priorities.

Power generation is provided by **Roll-Out Solar Arrays (ROSA)**, regulated through **Maximum Power Point Tracking (MPPT)** to maximize energy extraction across changing illumination conditions and incidence angles. The arrays are mounted on **four Solar Array Drive Assemblies (SADA)** enabling $\pm 180^\circ$ rotation, improving power availability and reducing sensitivity to attitude constraints and Sun-spacecraft geometry. Electrical power is regulated and distributed through a **dual-bus architecture** to accommodate heterogeneous loads: a **regulated 100 V bus** dedicated to the Hall thruster chain (via a PPU for voltage matching), and an **unreg-**

ulated 50 V bus feeding platform and payload loads through DC–DC conversion. Energy storage is based on **Li-ion secondary battery**, selected to minimize mass and volume while supporting repeated cycling. To preserve end-of-life capability and limit degradation over the expected eclipse cycles (order of 40k - 55k cycles over the minimum lifetime), the operational strategy constrains the battery **Depth of Discharge** to **30%**.

The EPS is dimensioned using two power-driving mission conditions: the **relocation mode**, representative of peak power demand (2 kW), and the **eclipse mode**, representative of battery-driven operation (0.37 kW). The sizing logic adopted allows the EPS to sustain the battery driver **maximum eclipse period** (188 min) in the operational environment, and the **maximum eclipse-to-sunlit ratio** case as the solar array sizing driver. The resulting configuration adopts four ROSA wings with an area of about 9.3 m² each, with a total array mass of ~ 119 kg) and a secondary battery sized at end-of-life to about 8.0 kWh (mass ~ 85 kg, volume ~ 89 L).

EPS compatibility with the low-thrust transfer strategy is ensured through an **EPS-compliant control logic** embedded on top of the guidance law. During transfers, the simulator monitors solar array power, mode-dependent load demand, and battery State-of-Charge. When instantaneous power gaps occur, battery support is allowed if the SoC remains above safety thresholds; otherwise, thrust is inhibited and the spacecraft transitions to a temporary low-power coast/safe configuration to recover energy before resuming thrust. This mechanism reshapes the guidance-level thrust request into an executed thrust/coast schedule that remains compatible with EPS constraints. A Monte Carlo campaign ($N = 1000$ transfers) confirms the feasibility of the approach and its robustness, including under a representative **single-SADA stuck fault** injection scenario.

7.8 Thermal Control Subsystem

The **PHAROS-M** Thermal Control Subsystem (TCS) ensures the survival and nominal operation of all spacecraft units. The design is verified through global Worst Hot and Worst Cold analyses performed via **ESATAN-TMS**.

The architecture implements a **dual-purpose structural philosophy** centered on the central cylinder. First, it acts as a geometric separator, dividing the spacecraft into internal thermally decoupled components (requiring dedicated thermal control) and an external zone where components are conductively coupled.

Second, for this external section, the cylinder serves as a high-inertia thermal dampener, absorbing transient fluctuations to equalize the overall temperature gradient.

To strictly enforce this strategy, specific hardware solutions are applied. **Multi-Layer Insulation** (20-layer Aluminized Kapton) provides the necessary radiative isolation from the external environment, preventing thermal leakage that would compromise the damping effect. Conversely, a 100 W network of film heaters is integrated to actively inject energy during eclipse phases, preventing the structure from stabilizing at non-operative cold temperatures. Furthermore, **fiberglass thermal washers** decouple the internal units from the structure, while **flexible copper thermal straps** create low-resistance paths to exchange heat with the central cylinder. The passive baseline is completed by four white-painted radiators, specifically distributed to dissipate thermal loads from the Hall Thruster (2 units) and electronics (2 units).

Numerical results confirm the validity of this approach, ensuring full compliance within operational ranges and a safety margin of 15 K.

7.9 Robotics and Mechanisms

PHAROS-M is equipped with a 7 degree-of-freedom **robotic arm** to enable repair and relocation operations. The base of the robotic arm is placed on the servicing side of the spacecraft, where it can cooperate with the 2+1 degree-of-freedom **clamping mechanism**. In both of the servicing tasks, the robotic arm executes berthing and capture, and subsequently delivers the client spacecraft to the clamping mechanism which, during repair, frees up the robotic arm to carry out tasks that require high dexterity; and during relocation, relieves the robotic arm of thrust loads with a client captured to the shorter, body-aligned clamping mechanism.

Due to the uncertainty of client spacecraft, and thereby their dimensions, additional degrees of freedom of the clamping mechanism were added to the clamping mechanism such that centers of mass of PHAROS-M and the client can be aligned. An additional benefit is that this allows the **manipulation of client** during repair to provide improved access and reachability to the robotic arm.

To aid in identifying client state and allow inspection for servicing tasks, the robotic arm is equipped with various sensors. Placed near the end effector are the **time-of-flight** and the **stereo visible range cameras**, which aid in berthing, capture and repair, and are active at different ranges from target. A **force-torque sen-**

sor enables the robotic arm to detect a successful capture and non-nominalities during operation. An off-the-shelf **gripper** for both the robotic arm and the clamping mass is selected, with modification to enable the capture of multiple **launch adapter ring (LAR)** sizes. The cross-communication between the robotic arms and the GNC subsystem are required only during berthing and transfer of client from the arm to the clamping mechanism, as the stack is otherwise **rigidized** by the clamping mechanism.

7.10 Configuration and Structures

The **configuration** of **PHAROS-M** is designed to meet the requirements imposed by the other subsystems in the best way, to provide safe and secure attachment of both external and internal components, and to ensure a rigid, strong, and lightweight structure. The satellite will be mounted on top of **MANTIS**, protected from the launcher and space environment by their structure, and the entire stack will be installed longitudinally inside the launcher. This is why the **bottom face** hosts the **ring interface**, the primary, and part of the secondary propulsion. On the **top face**, the servicing features are arranged, with the idea of dedicating the entire face to the capture and servicing of the clients.

The shape of the bus is an **octagonal prism** with an octagonal **truncated cone** upper base. The need of mounting a lot of components on the external faces requires having a polygonal-shaped based; while the possibility to have, especially during Maneuver Mode, four solar panels to maximize the illumination of the cells involves the choice of the octagonal base. The axial direction, i.e. the longitudinal one, is the z -axis, while x and y are the two in-plane axes, as can be seen in the images presented in section 10. Moreover, the need of **GNC** sensors to be oriented towards the client to perform **Rendezvous and Proximity Operations**, without being damaged or compromised during the docking/berthing operations, nor by the Robotic Arm, dictates the choice of a truncated upper shape.

The distribution of **external components** is the result of different optimization processes in order to find the best combination in terms of inertia moments, CoG offsets, attitude and communication strategies, mutual impingement and clearances, and satisfactory servicing conditions. In addition, 16 **resistojets** are arranged as follows: 8 on the lateral faces and 8 on the bottom, to better exploit the distance from the satellite's CoG to generate torque in all the directions, to allow **PHAROS-M** to be always maneuverable.

The placement of **internal components** is dic-

tated instead by the pursuit of simplicity and the global equilibrium of the structure.

From a **structural** point of view, a **central cylinder**, whose diameter matches the ring adapter, constitutes the satellite's backbone through which the launch and transfer loads can propagate. Moreover, four **vertical ribs** are arranged symmetrically around the cylinder, providing a strong load path to distribute the loads with the octagonal bus. Finally, to place the N2H4 tanks and the RWs, two **horizontal decks** are designed inside the central cylinder.

Given the tight **constraints** in terms of **mass** imposed by the carrier, and the need of the structure to be quite **stiff** to avoid possible resonance dynamics with MANTIS, the satellite's structure is entirely made of sandwich panels, with the exception of the ring adapter that is in Aluminum 7075-T6. This choice allows **PHAROS-M** to better deal with concentrated loads at the carrier's interface and to reach the desired level of stiffness, while saving mass.

8 COST, RISK & PROGR.

The programmatic and economic framework for **PHAROS-M** is developed in adherence to **ECSS standards**, ensuring end-to-end traceability across the Product Tree, Risk Register, Model Philosophy, and the resulting Cost and Schedule baselines. **Risk Management** follows two iterations: a level-zero analysis of an unmitigated product tree isolates the dominant drivers and identifies the **Servicing Payloads** and the **GNC architecture** as the critical path; a second iteration embeds mitigations directly into the final product tree via targeted redundancies and dedicated test models. This profile supports the adoption of a **Hybrid Model Philosophy**, with qualification models assigned to high-novelty subsystems to reduce development risk and the flight unit following a protoflight workflow to optimise verification; the associated **non-flight models** are explicitly included in the product tree to preserve full budget traceability. Programmatically, the mission targets a **launch window** in **April 2033**, and the schedule is structured around ECSS review gates to freeze the technical baseline before procurement and AIV. The **Total Lifecycle Cost** is estimated at **1.2 billion euros** from Phase A through disposal, based on a bottom-up aggregation combining similarity with deep-space platforms and parametric sizing; uncertainty is captured through TRL-based contingencies and residual risk margins. A preliminary **S-curve** converts the Gantt logic into a funding profile, with peak expenditure in **Phases D–E** driven by flight hardware procure-

ment and the Launch and Transport segment. Finally, **economic viability** is assessed by comparing the **PHAROS-M** total cost against the **replacement cost** of Martian assets: scaling Earth on-orbit servicing pricing to Mars (interplanetary logistics and autonomy) indicates a 4–6× increase relative to LEO, but over a 12-year lifetime and **5–6 clients** the cost is amortised across users, defining conditions where **servicing becomes preferable** to developing and launching new Mars orbiters (700–800 M\$ per unit) and positioning **PHAROS-M** as an enabler of a circular economic model for the Martian frontier.

9 CONCLUSION

PHAROS-M establishes a Mars-orbit servicing hub concept enabling repeated, autonomous **rendezvous, capture, and servicing** campaigns across a representative orbital cluster. The mission combines relay-supported operations, low-thrust transfers consistent with eclipse constraints, robust **GNC** for proximity operations, and a servicing payload capable of **refueling** and **repair-oriented** interventions. This framework supports **operational continuity** for critical Mars assets and improves sustainability by reducing reliance on replacement launches.

A key system-level outcome is that the **carrier mass constraints** cannot be respected under the current baseline. Serving an **elevated number of clients**, together with the **high autonomy level** and capability envelope required for deep-space servicing, drives multiple subsystems to high mass and volume. This is further amplified by the conservative **Mars environment** design approach, requiring a significant system-level margin, including **MAR-MAS-040 (+30% on the nominal dry mass at launch)**, pushing the spacecraft beyond the carrier allocation.

Future work will prioritise **mass closure** and **architecture convergence**: refining the client set and campaign assumptions; rebalancing autonomy and redundancy through mode criticality; consolidating the sensing and robotics chain while preserving safety and docking performance; improving packaging and stowage efficiency; and reducing mass uncertainty through higher-fidelity environmental modelling and verification-driven budgeting. In parallel, the ConOps will be matured by formalising eclipse and Sun-geometry inhibition rules and sharpening authorization gates as a function of relay availability and client constraints.

10 PHAROS-M CONFIGURATION

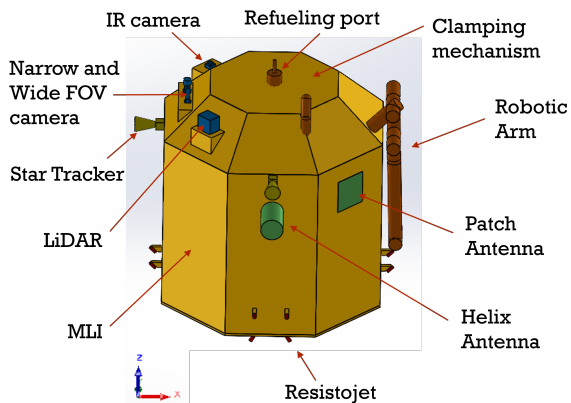


Figure 3: Octagonal bus with surface components shown.

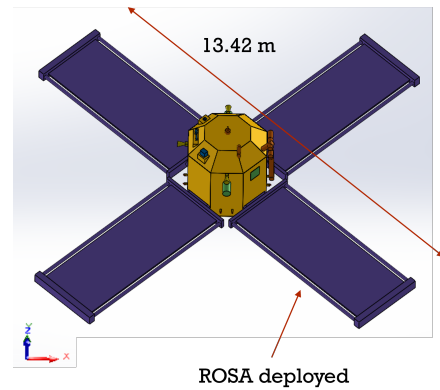


Figure 4: External isometric view (deployed)

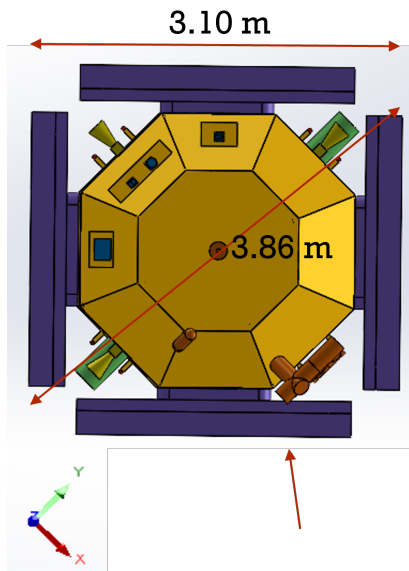


Figure 5: External top view (stowed)

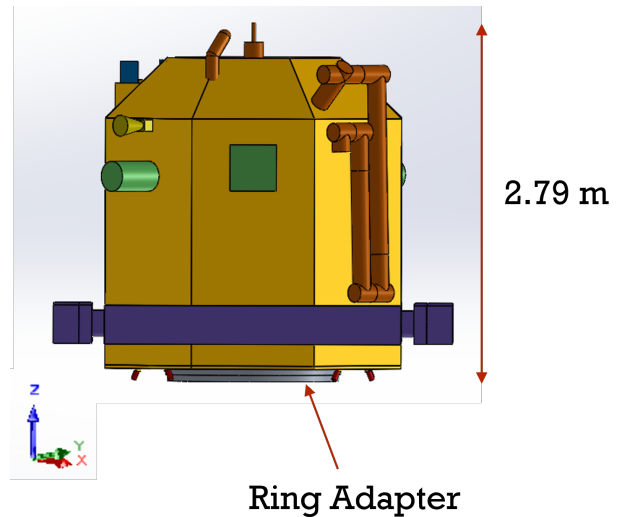


Figure 6: External side view (stowed)

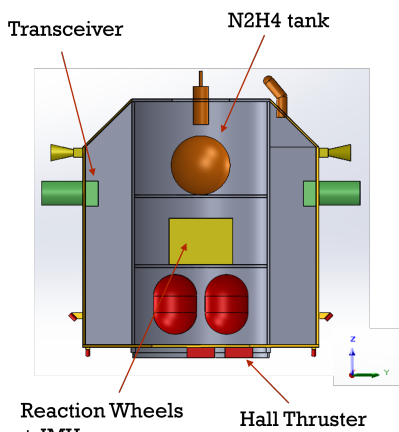


Figure 7: Section view: front

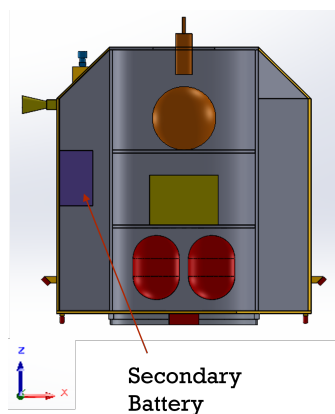


Figure 8: Section view: left

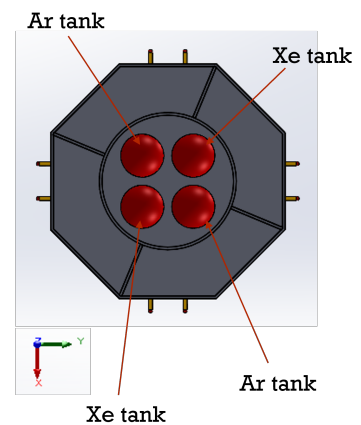


Figure 9: Section view: top